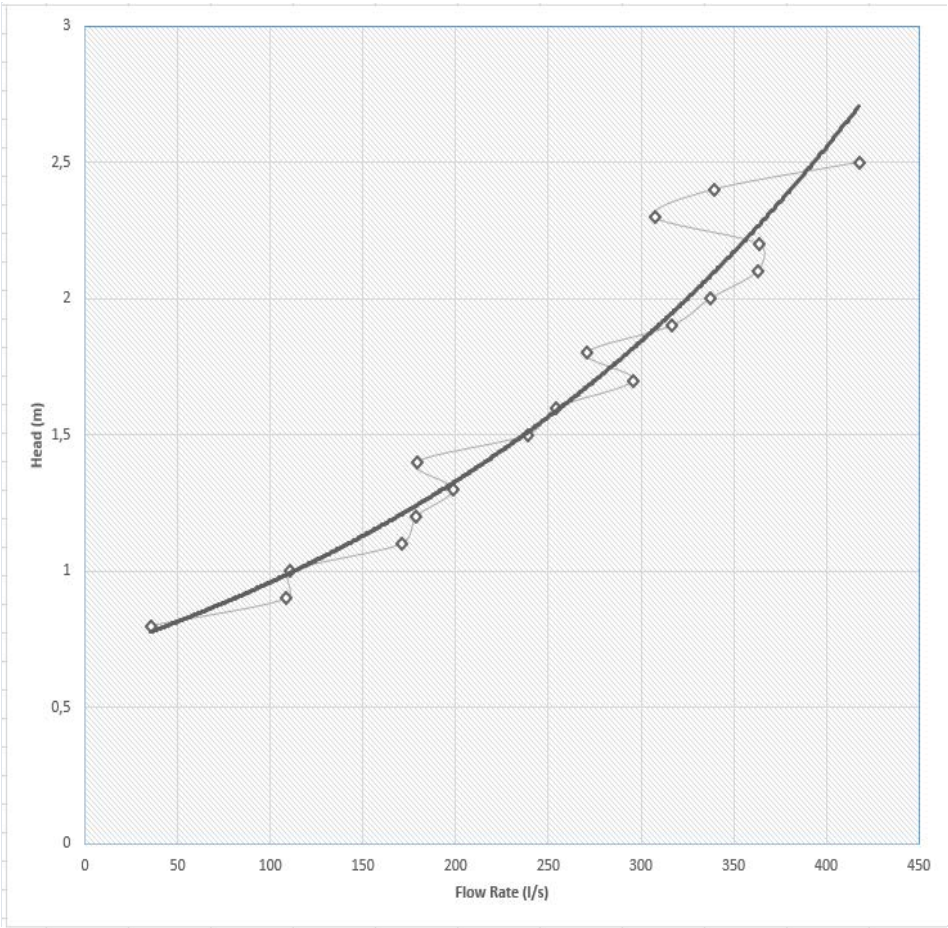
114mm LR3 Radial Type Vortex Flow Control

Client Name:		Date:	27/09/2023
Project Name:	Leeds	Job Reference:	



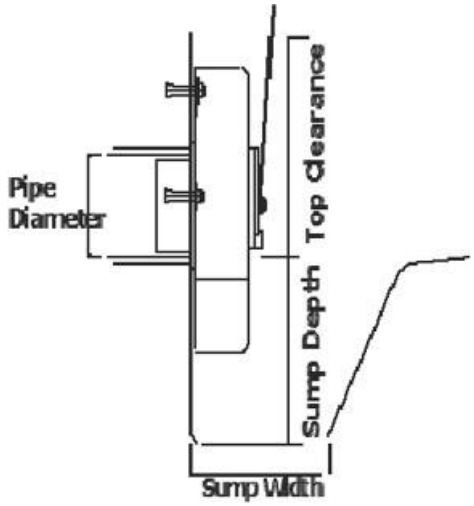
Head (m)	Flow l/s
0	0.000
0.1	35,374
0.2	108,637
0.3	110,668
0.4	170,617
0.5	178,576
0.6	198,608
0.7	178,967
0.8	238,616
0.9	254,393
1	295,748
1.1	270,715
1.2	316,443
1.3	337,382
1.4	362,790
1.5	363,732
1.6	307,286
1.7	339,229
1.8	417,625
1.9	334,797
2	339,358
2.1	376,191
2.2	403,7868

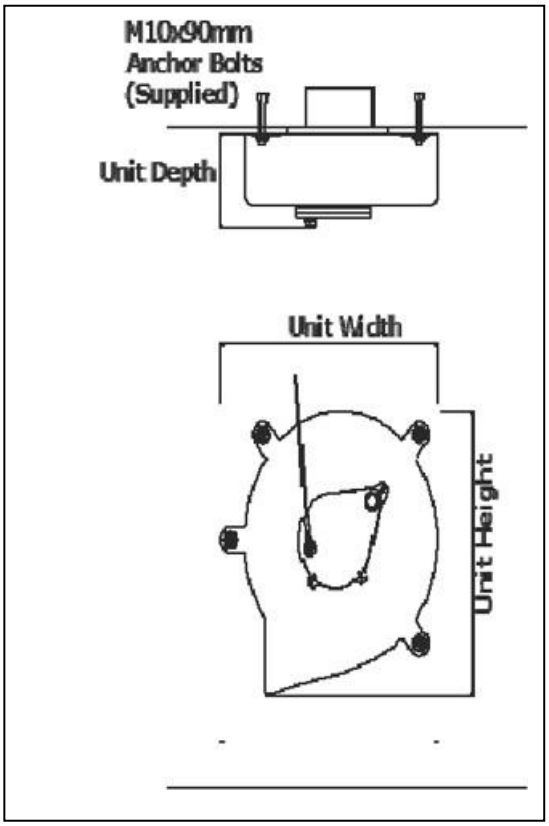
Design Flow:	235 l/s	Flush Flow:	180 l/s
Design Head:	0.77m	At Head:	0.510m
Minimum Pipe:	114mm	Kickback Flow:	179.20 l/s
Sump Depth:	380mm	At Head:	0.695m

Note: Surface Water Only

Client Name:		Date:	14/02/2022
Project Name:	Leeds	Job Reference:	

SuDS Vortex Flow Control Unit
Installation Guidelines:





Unit Outlet Dia:	114mm	Sump Depth (Min):	380mm
Unit Height:	215.19mm	Sump Width (Min):	513mm
Unit Width:	387.94mm	Pipe Diameter (Min):	114mm
Unit Depth:	170mm	Top Clearance (Min):	502mm

Note: Surface Water Only

Product Information and FAQ's

DESIGN:

All of our Vortex Flow Control Units are individually designed to meet the client's optimum performance criteria.

Will our Vortex Flow Control's Block?
The relatively large cross-sectional areas and high internal velocities greatly reduce the risk of blockage.

How Long Will They Last?
SuDS Enviro Vortex Flow Control Units have no moving parts to wear or fail. Made from stainless steel plate they will resist scour, degradation and chemical attack. A Vortex Flow Control should outlast the sewer in which it is installed.

MANUFACTURE:

SuDS Enviro Vortex Flow Control Units are manufactured to a high standard.

• 304 Grade Stainless Steel as standard. Other material optional.
• Plate thickness 3mm as standard. Thicker plate used where required.
• Pulse arc Tungsten Inert Gas Welding. All seams continuous for maximum strength.
• Emergency Bypass Door fitted as standard to allow drain down and rodding access.
• Externally regulated ISO 9001:2008 Quality Assurance System

INSTALLATION:

All of SuDS Enviro's Vortex Flow Controls are purpose built to suit the specific site and to fit easily into the drainage infrastructure. Units can be specially designed for retrofitting or for installation in difficult spaces.

MAINTENANCE:

Vortex Flow Control units themselves require no routine maintenance. Inspection chambers require inspection & maintenance in line with current practice.

WHAT DESIGN INFORMATION DO WE NEED?

We require the following basic information to specify a Vortex Flow Control for you.
1) The design Flow - *Maximum Discharge*
2) The design Head - *Invert to Top Water Level*
From this information we will size and design the Vortex Flow Control to meet the design criteria and to suit the proposed infrastructure.

SPECIFICATION

Includes: -
1) Data Sheet showing the Head/Discharge Characteristic, both graphically and numerically.

2) Installation Guide showing the outline dimensions of the specified Vortex Flow Control and the method of fixing.

3) Quotation covering the design manufacture.

SUMMARY:

In summary, the simulation of the new design has revealed a flushing flow rate that is 2 to 4 times better than what is required to meet our needs. The design boasts a flow rate of 35,374 liters per second at a 0.1 meter head height and 504,908.6 liters per second at a 2.5 meter head height. Furthermore, the LR3 Radial Flow Control Device incorporates a buoyancy-driven emergency door opening system, which serves to flush accumulated water from both the tank and the street above in the event of any blockages occurring at the main exit. Notably, this system's operation creates a new vortex, potentially aiding in the removal of blockages and facilitating self-maintenance.